Noncommuting Hermitian Matrix-Polynomial Filters Can Have an Exponential Tangential Advantage

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Abstract

Scalar Chebyshev filtering treats every vector in a block Krylov iterate with the same polynomial, while simultaneously diagonalizable matrix coefficients amount to independent scalar filters after a fixed channel rotation. We study the larger class of Hermitian matrix-polynomial filters under tangential pass constraints. An exact full-spark quadratic certificate already separates noncommuting from commuting symmetric coefficients on a full stopband. Our main result gives an asymptotic separation: at degree N and channel dimension $d = N^2$, there are distinct pass points and full-spark channel signatures for which a symmetric noncommuting filter has stopband leakage $O(e^{-cN})$, whereas every feasible pairwise-commuting symmetric filter is identically the identity and has leakage one. We give an explicit full-spark perturbation and a block-Krylov error interpretation. Numerical semidefinite programs support the finite-dimensional mechanism, and a controlled Hermitian block iteration tests the resulting subspace error, but neither is used in the proof. This draft makes no claim that tangential matrix interpolation or matrix-valued Krylov theory is new; priority of the specific minimax separation remains under audit.

1 Problem and contribution

Let

$$P(x) = \sum_{k=0}^{N} C_k x^k, \qquad C_k \in \mathbb{R}^{d\times d},$$

and let $I_- = [-1, 0]$ be a stopband. Given distinct pass points $\lambda_j > 0$ and nonzero channel signatures $v_j \in \mathbb{R}^d$, impose

$$P(\lambda_j)v_j = v_j, \qquad 1 \le j \le M. \tag{1}$$

The leakage is

$$\rho(P) = \sup_{x\in I_-} \|P(x)\|_{\mathrm{op}}. \tag{2}$$

The coefficients will always be real symmetric. They need not commute.

The comparator is the pairwise-commuting subclass. Such coefficients admit a common orthogonal eigenbasis, so a fixed channel rotation turns P into a diagonal collection of scalar filters. The question is whether allowing the channel mixing to depend on the polynomial degree can reduce (2) under the same pass constraints.

Our answer is exponentially affirmative in an existence sense. The input signatures are full spark: every d of them span $\mathbb{R}^d$. This property makes the commuting obstruction exact, while an arbitrarily small, explicitly chosen perturbation of a coordinate construction retains a noncommuting interpolant with exponentially small leakage.

2 Main separation theorem

Theorem 2.1 (Exponential tangential separation)**.** *There are absolute constants $C, c > 0$ and $N_0 \in \mathbb{N}$ such that for every $N \geq N_0$, with $d = N^2$ and $M = d + N$, one can choose distinct rational pass points $\lambda_1, \ldots, \lambda_M \in [1, 7/2]$ and full-spark unit vectors $v_1, \ldots, v_M \in \mathbb{R}^d$ with the following properties.*

1. *There is a degree-N polynomial P_N with real symmetric coefficient matrices satisfying* (1) *and*
$$\rho(P_N) \leq Ce^{-cN}.$$
2. *If a degree-N polynomial Q_N has pairwise-commuting real symmetric coefficient matrices and satisfies the same tangential constraints, then $Q_N(x) = I_d$ for every x. In particular, $\rho(Q_N) = 1$.*

For all sufficiently large N, the coefficient matrices of P_N cannot commute pairwise.

We split the proof into an obstruction, a small-leakage base construction, and a stable full-spark perturbation.

2.1 The commuting obstruction

Lemma 2.2. *Let $M \geq d + N$, let the λ_j be distinct, and let the $v_j \in \mathbb{R}^d$ be full spark. If $Q(x) = \sum_{k=0}^{N} D_k x^k$ has pairwise-commuting real symmetric coefficients and $Q(\lambda_j)v_j = v_j$ for every j, then $Q = I_d$.*

Proof. The matrices D_k have a common orthonormal eigenbasis $u_1, \ldots, u_d$. Write $q_r(x)$ for the scalar degree-N polynomial induced by $Q(x)$ on u_r. Taking the u_r component of the jth interpolation constraint gives
$$(q_r(\lambda_j) - 1)\langle u_r, v_j \rangle = 0.$$
A nonzero vector can be orthogonal to at most $d - 1$ of the full-spark targets: otherwise d targets would lie in the $(d-1)$-dimensional hyperplane $u_r^\perp$ and could not span $\mathbb{R}^d$. Thus at least $M - d + 1 \geq N + 1$ distinct values of λ_j are roots of $q_r - 1$. Consequently $q_r = 1$. This holds for every r, so $Q = I_d$. □

2.2 A diagonal base filter

Let $d = N^2$. Partition $M = d + N$ nodes into d coordinate groups: the first N groups contain two nodes each and the remaining groups contain one. One concrete choice, with $1 \leq r \leq N$ and $N < r \leq d$, is

$$a_r = 1 + \frac{r}{10N}, \qquad b_r = 2 + \frac{r}{10N}, \tag{3}$$
$$c_r = 3 + \frac{r - N}{2d}. \tag{4}$$

These nodes are distinct, lie in $[1, 7/2]$, and have mutual separation at least $(2N^2)^{-1}$.

Write T_m for the Chebyshev polynomial of the first kind and set $\phi(x) = 2x + 1$, which maps I_- onto $[-1, 1]$. For a one-node group with node λ, define

$$q_\lambda(x) = \frac{T_N(\phi(x))}{T_N(\phi(\lambda))}. \tag{5}$$

For a two-node group with nodes $a \neq b$, define

$$q_{a,b}(x) = \frac{x - b}{a - b}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(a))} + \frac{x - a}{b - a}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(b))}. \tag{6}$$

These polynomials equal one at their assigned node or nodes. Put them on the diagonal of $P_0(x)$ according to the coordinate partition.

Lemma 2.3. *Let*

$$\eta_0 = \operatorname{arcosh}(3), \qquad \eta_1 = \operatorname{arcosh}(8), \qquad \Delta\eta = \eta_1 - \eta_0.$$

Then

$$\sup_{x \in I_-} \|P_0(x)\|_{\mathrm{op}} \le \frac{52}{5} e^{\eta_0} e^{-\eta_0 N}.$$

Moreover, at every pass node $\lambda_j \in [1, 7/2]$, $\|P_0(\lambda_j)\|_{\mathrm{op}} \le 10e^{N\Delta\eta}$ uniformly in the group labels.

Proof. For $x \in I_-$, $|T_m(\phi(x))| \le 1$. For $\lambda \ge 1$,

$$T_m(\phi(\lambda)) = \cosh(m \operatorname{arcosh}(2\lambda + 1)) \ge \tfrac{1}{2} e^{m\eta_0}, \quad \eta_0 = \operatorname{arcosh}(3).$$

The one-point stopband bound is $2e^{-N\eta_0}$. In (3), $b_r - a_r = 1$. On I_-, the sum of the absolute Lagrange factors in (6) is at most $31/10 + 21/10 = 26/5$. The two-point bound is therefore $(52/5)e^{-(N-1)\eta_0}$.

At a pass node, $2\lambda+1 \le 8$, so a Chebyshev ratio is at most $2e^{N\Delta\eta}$. For the paired construction, the sum of the absolute Lagrange factors is at most five. This gives the stated pass-node bound. The operator norm of a diagonal polynomial is its largest entry magnitude. □

2.3 An explicit full-spark perturbation

Index all pass nodes by $j = 1, \dots, M$, let $g(j)$ be the coordinate group of node j, and set $t_j = j/(M + 1)$. Define

$$w_j = (1, t_j, \dots, t_j^{d-1})^\top, \quad \varepsilon_N = e^{-N^3}, \quad \widetilde{v}_j = e_{g(j)} + \varepsilon_N w_j, \quad v_j = \frac{\widetilde{v}_j}{\|\widetilde{v}_j\|_2}. \tag{7}$$

Lemma 2.4. *The vectors in* (7) *are full spark.*

Proof. Fix any d indices. The determinant with the corresponding unnormalized vectors as columns is a polynomial in ε_N with rational coefficients. Its coefficient at degree d is the Vandermonde determinant of the corresponding w_j, which is nonzero because their t_j are distinct. Thus the determinant polynomial is not zero. By the Lindemann–Weierstrass theorem, e^{-N^3} is transcendental, so it cannot be a root of a nonzero polynomial with rational coefficients. Column normalization does not change whether the determinant vanishes. □

Lemma 2.5 (Stable symmetric interpolation)**.** *For all sufficiently large N, there is a real-symmetric coefficient polynomial P_N satisfying $P_N(\lambda_j)\widetilde{v}_j = \widetilde{v}_j$ and*

$$\sup_{x \in I_-} \|P_N(x) - P_0(x)\|_{\mathrm{op}} \le 64152\, N^9 \exp(-N^3 + N\Delta\eta).$$

Proof. Let $\mathcal{S}_N$ be the finite-dimensional space of degree-at-most-N polynomials with real symmetric coefficients, and define

$$L_\varepsilon : \mathcal{S}_N \longrightarrow (\mathbb{R}^d)^M, \qquad L_\varepsilon(P) = \big(P(\lambda_j)(e_{g(j)} + \varepsilon w_j)\big)_{j=1}^M.$$

At $\varepsilon = 0$, this map is onto. Indeed, for an off-diagonal entry $p_{rs} = p_{sr}$, arbitrary output data prescribe values at the union of nodes assigned to coordinates r and s. That union contains at most four nodes. For $N \ge 3$, ordinary Lagrange interpolation is onto on those nodes. A diagonal entry sees at most two nodes.

Equip the output space with the maximum Euclidean block norm and $\mathcal{S}_N$ with

$$\|P\|_X = \max\left\{\sup_{x\in I_-}\|P(x)\|_{\mathrm{op}}, \max_j \|P(\lambda_j)\|_{\mathrm{op}}\right\}.$$

The entrywise construction gives a right inverse R_N. For $N \geq 5$, the node separation is at least $(2N^2)^{-1}$, while any two points in the stop/pass domain are at distance at most $9/2$. A Lagrange basis polynomial on at most four nodes is bounded by $(9N^2)^3$. Summing at most four basis polynomials and using $\|A\|_{\mathrm{op}} \leq d\max_{r,s}|A_{rs}|$ gives

$$\|R_N\|_{Y\to X} \leq 4(9N^2)^3 d = 2916N^8. \tag{8}$$

Write $L_\varepsilon = L_0 + \varepsilon K_N$. Since $\|w_j\|_2 \leq \sqrt{d} = N$, $\|K_N\|_{X\to Y} \leq N$. Hence

$$L_\varepsilon R_N = I + E_N, \qquad \|E_N\| \leq 2916\varepsilon N^9.$$

For $\varepsilon = \varepsilon_N$ and large N, the inverse $(I+E_N)^{-1}$ exists and has norm at most two.

Since $L_0(P_0) = (e_{g(j)})_j$, the residual at the perturbed targets is

$$r_N = (\widetilde{v}_j)_j - L_{\varepsilon_N}(P_0) = \varepsilon_N\big((I - P_0(\lambda_j))w_j\big)_j.$$

By Lemma 2.3,

$$\|r_N\|_Y \leq \varepsilon_N N(1 + 10e^{N\Delta\eta}) \leq 11\varepsilon_N N e^{N\Delta\eta}.$$

Set

$$\Delta P_N = R_N(I+E_N)^{-1}r_N, \qquad P_N = P_0 + \Delta P_N.$$

Then $L_{\varepsilon_N}(P_N) = (\widetilde{v}_j)_j$. Using (8) and $\|(I+E_N)^{-1}\| \leq 2$ gives

$$\|\Delta P_N\|_X \leq 2(2916N^8)(11\varepsilon_N N e^{N\Delta\eta}) = 64152N^9 e^{-N^3+N\Delta\eta}.$$

Every coefficient remains real symmetric because R_N maps into $\mathcal{S}_N$. □

Proof of Theorem 2.1. The normalized and unnormalized interpolation constraints are equivalent. Combine Lemmas 2.3, 2.4, and 2.5. Choose $N_0 \geq 5$ so that $2916e^{-N^3}N^9 \leq 1/2$ and the correction in Lemma 2.5 is at most $(52/5)e^{\eta_0}e^{-\eta_0 N}$ for $N \geq N_0$. The first conclusion holds with

$$C = \frac{104}{5}e^{\eta_0}, \qquad c = \eta_0.$$

The second conclusion is Lemma 2.2. For large N, the first leakage is below one, so the constructed coefficients cannot commute pairwise. □

3 Block-Krylov meaning

The theorem has a direct interpretation for block polynomial filtering. Let $H = \sum_j \lambda_j q_j q_j^\top$ be real symmetric and let $B \in \mathbb{R}^{n\times d}$ be a starting block. For $P(x) = \sum_{k=0}^N C_k x^k$, define the right-coefficient block filter

$$\mathfrak{F}_P(H,B) = \sum_{k=0}^{N} H^k B C_k. \tag{9}$$

Its jth spectral row is

$$q_j^\top \mathfrak{F}_P(H,B) = (q_j^\top B)P(\lambda_j). \tag{10}$$

Proposition 3.1 (Pass preservation and stop contraction)**.** *Let $\mathcal{T}$ and $\mathcal{S}$ index target and stop eigenpairs. Assume the rows $b_j^\top = q_j^\top B$ obey*

$$b_j^\top P(\lambda_j) = b_j^\top \quad (j \in \mathcal{T}), \qquad \lambda_j \in I_- \quad (j \in \mathcal{S}).$$

Then the target spectral component is preserved exactly, and

$$\|Q_{\mathcal{S}}^\top \mathfrak{F}_P(H,B)\|_F \le \rho(P) \|Q_{\mathcal{S}}^\top B\|_F,$$

where $Q_{\mathcal{S}}$ has columns q_j, $j \in \mathcal{S}$. If the C_k are pairwise-commuting symmetric matrices, one fixed orthogonal channel rotation reduces (9) *to d independent scalar polynomial filters.*

Proof. The pass statement follows from (10). For the stop part,

$$\|Q_{\mathcal{S}}^\top \mathfrak{F}_P(H,B)\|_F^2 = \sum_{j\in\mathcal{S}} \|b_j^\top P(\lambda_j)\|_2^2 \le \rho(P)^2 \sum_{j\in\mathcal{S}} \|b_j\|_2^2.$$

For the final claim, simultaneously diagonalize all $C_k = U\Lambda_k U^\top$. Then the columns of BU are acted on by the scalar polynomials whose coefficients are the diagonal entries of the Λ_k. $\square$

Because every coefficient in Theorem 2.1 is symmetric, its column constraints transpose to the row constraints in Theorem 3.1. The theorem therefore gives block spectral instances with exponentially small stop contraction that no fixed-basis collection of scalar pass-preserving filters can reproduce. This is an approximation-theoretic separation; an end-to-end runtime advantage also depends on filter construction, recurrence stability, orthogonalization, and the cost model.

4 Exact finite witnesses

4.1 A full-spark quadratic certificate

The asymptotic mechanism already occurs with rational data at degree two. Take

$$(\lambda_1,\lambda_2,\lambda_3,\lambda_4) = \left(\tfrac{1}{2}, 1, \tfrac{3}{2}, 2\right), \qquad (v_1,v_2,v_3,v_4) = \left(\begin{pmatrix}1\\-2\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}\right).$$

Every pair of targets is linearly independent, hence the tuple is full spark in $\mathbb{R}^2$. Set $P(x) = C_0 + xC_1 + x^2C_2$, with

$$C_0 = \begin{pmatrix}2/5 & -3/16\\ -3/16 & 29/32\end{pmatrix}, \quad C_1 = \begin{pmatrix}15/16 & 3/20\\ 3/20 & 3/32\end{pmatrix}, \quad C_2 = \begin{pmatrix}-3/8 & 0\\ 0 & -3/80\end{pmatrix}.$$

Direct substitution verifies $P(\lambda_j)v_j = v_j$ for all four targets, and

$$[C_0, C_1] = \begin{pmatrix}0 & 1053/12800\\ -1053/12800 & 0\end{pmatrix} \neq 0.$$

Proposition 4.1 (Exact quadratic gap)**.** *The polynomial above obeys*

$$\sup_{x\in[-1,0]} \|P(x)\|_{\mathrm{op}} \le \frac{301}{304} < 1.$$

Every pairwise-commuting real-symmetric quadratic polynomial satisfying the same four tangential constraints is identically I_2 and has leakage one.

Proof. For the upper bound, put $x = t - 1$ and express the leading principal minor and determinant of $I_2 \pm P(t-1)$ in the Bernstein basis on $t \in [0,1]$. The leading-minor coefficients are

$$\left(\tfrac{153}{80}, \tfrac{171}{160}, \tfrac{3}{5}\right), \qquad \left(\tfrac{7}{80}, \tfrac{149}{160}, \tfrac{7}{5}\right),$$

and the determinant coefficients are, respectively,

$$\left(\tfrac{81}{256}, \tfrac{1701}{10240}, \tfrac{219}{2560}, \tfrac{441}{10240}, \tfrac{27}{1280}\right),$$
$$\left(\tfrac{53}{1280}, \tfrac{8389}{10240}, \tfrac{783}{512}, \tfrac{21913}{10240}, \tfrac{3371}{1280}\right).$$

They are all positive, so both matrices are positive definite throughout the stopband. Their trace Bernstein coefficients give the upper bounds 171/80 and 529/160. For a positive 2 by 2 matrix S, $\lambda_{\min}(S) \geq \det(S)/\operatorname{tr}(S)$. Consequently

$$I_2 - P(x) \succeq \frac{3}{304} I_2, \qquad I_2 + P(x) \succeq \frac{53}{4232} I_2 \succeq \frac{3}{304} I_2,$$

which proves the norm bound.

For the commuting class, apply Lemma 2.2 with $M = 4 = d + N$. Every scalar common-eigenbasis entry has three distinct roots of $q - 1$ and is identically one. $\square$

The exact-arithmetic replay `verification/verify_quadratic_filter_witness.py` checks all constraints, target minors, commutators and Bernstein coefficients without a numerical solver.

4.2 An affine two-constraint precursor

For degree one and $d = 2$, take pass data

$$\lambda_1 = \tfrac{1}{2}, \quad v_1 = (1,0)^\top, \qquad \lambda_2 = 1, \quad v_2 = (1,1)^\top,$$

and $P(x) = A + xB$ with

$$A = \begin{pmatrix} 4/5 & 1/5 \\ 1/5 & 3/10 \end{pmatrix}, \qquad B = \begin{pmatrix} 2/5 & -2/5 \\ -2/5 & 9/10 \end{pmatrix}.$$

The constraints hold exactly and

$$[A, B] = \begin{pmatrix} 0 & -1/10 \\ 1/10 & 0 \end{pmatrix}.$$

Convexity of $x \mapsto \|A + xB\|_{\mathrm{op}}$ puts its maximum over I_- at an endpoint. Direct diagonalization gives

$$\rho(P) = \max\{\|A\|_{\mathrm{op}}, \|A - B\|_{\mathrm{op}}\} = \frac{1 + \sqrt{61}}{10} < 0.882.$$

If A and B commute, their common eigenbasis contains a direction with nonzero overlap with both nonorthogonal, nonparallel targets. Its scalar affine entry equals one at two points and hence everywhere. The commuting leakage is at least one.

5 Numerical stress test

The repository script `research/route_c_scaling_pilot.py` solves a grid SDP with symmetric coefficient variables. For degree 3, dimension 9, 12 perturbed targets and seed 7, the results are:

Perturbation	SDP objective	Validation maximum	Commutator norm
0.001	0.2983490	0.2983682	0.466925
0.010	0.3199793	0.3200281	0.391179
0.030	0.3977121	0.3977588	0.708604

The optimization uses 81 stopband points and validation uses 4001 independent points. Clarabel labels these solves `optimal_inaccurate`; they are diagnostics, not proof. The exact witnesses and Theorem 2.1 do not depend on solver output.

5.1 Controlled block-iteration benchmark

The script `research/route_c_block_krylov_benchmark.py` embeds the quadratic certificate in a 256-dimensional Hermitian eigenproblem. The four pass eigenvalues are $1/2, 1, 3/2, 2$, the other 252 eigenvalues fill $[-1, 0]$, and the initial stop/pass Frobenius ratio is five. We repeatedly apply the degree-two matrix filter and compare it with

$$q(x) = \frac{8x^2 + 8x + 1}{7},$$

the scalar degree-two Chebyshev stopband filter normalized only at $x = 1/2$. Let $\sin\theta$ denote the largest principal-angle sine from the filtered block to the clean two-dimensional block defined by the four prescribed pass rows. The scalar pass error below is minimized over one global rescaling, so it does not merely report scalar amplitude growth.

Iterations	Matrix stop/pass	Matrix $\sin\theta$	Scalar pass error	Scalar $\sin\theta$
0	5.0000	0.9892	0.0000	0.9892
1	3.5776	0.9719	0.5616	0.5568
5	2.4352	0.9458	0.7832	0.9727
20	1.0318	0.7755	0.8018	0.9871
100	0.0443	0.0525	0.8018	0.9982

The matrix pass error stays below 6×10^{-16}. The scalar filter has stopband maximum $1/7$ and initially suppresses contamination much faster, but its pass multipliers are $(1, 17/7, 31/7, 7)$; iteration consequently rotates the clean block away from the prescribed one. The fair commuting comparator under all four exact tangential constraints is the identity and remains at stop/pass ratio 5 and $\sin\theta = 0.9892$. This controlled construction demonstrates the theorem's error mechanism, not a runtime advantage on a natural problem.

6 Relation to established work and open gates

Tangential matrix-polynomial interpolation is classical [2, 4]. Matrix-valued Nevanlinna–Pick theory handles supremum-norm analytic interpolation, including complexity constraints [3, 1]. Kuroiwa and Lindquist treat the bitangential problem with explicit McMillan-degree constraints [5]. Most directly, Stefanovski and Georgijević prove that real stable rational matrices can meet bitangential constraints while having arbitrarily small spectral norm on a proper frequency region [7]; their degree grows and they do not compare commuting and noncommuting Hermitian polynomial coefficients. Thus small regional norm under matrix tangential constraints is not itself our contribution. Minimax polynomial filtering and SDP formulations have a long numerical-linear-algebra history [8], and matrix-polynomial spaces arise naturally in block Krylov methods [6]. Accordingly, none of these ingredients is claimed as new.

The candidate novelty is therefore narrower than norm-constrained or degree-constrained tangential interpolation alone. Under a fixed ordinary degree, a Hermitian-coefficient tangential minimax problem on a real interval can exhibit an exponential separation between noncommuting coefficients and the full pairwise commuting subclass. Before submission, these gates remain:

1. compare Theorem 2.1 line by line with full texts on norm-constrained tangential interpolation, rather than relying on titles or abstracts;

2. test a naturally arising multichannel eigenproblem in which pass-row geometry is scientifically mandated, including adaptive and nonstationary scalar baselines beyond the controlled construction above.

Reproducibility. The source and research ledger are in the project repository. The repository links the exact rational replay, the block-iteration benchmark, its machine-readable output, and the artifact manifest with the PDF hash.

References

[1] Joseph A. Ball and Vladimir Bolotnikov. The bitangential matrix Nevanlinna–Pick interpolation problem revisited. *Linear Algebra and its Applications*, 516:122–188, 2017.

[2] Joseph A. Ball and Jeongook Kang. Matrix polynomial solutions of tangential Lagrange–Sylvester interpolation conditions of low McMillan degree. *Linear Algebra and its Applications*, 137–138:699–746, 1990.

[3] Anders Blomqvist, Anders Lindquist, and Ryozo Nagamune. Matrix-valued Nevanlinna–Pick interpolation with complexity constraint: an optimization approach. *IEEE Transactions on Automatic Control*, 48(12):2172–2190, 2003.

[4] Paul A. Fuhrmann. On tangential matrix interpolation. *Linear Algebra and its Applications*, 433(11–12):2018–2059, 2010.

[5] Yohei Kuroiwa and Anders Lindquist. Bi-tangential Nevanlinna–Pick interpolation with a complexity constraint. In *Proceedings of the 17th International Symposium on Mathematical Theory of Networks and Systems*, pages 2541–2560, Kyoto, Japan, 2006.

[6] Michele Rinelli and Raf Vandebril. Block Krylov subspaces and orthogonal matrix polynomials: a structural correspondence with applications to unitary matrices. 2026.

[7] Jovan Stefanovski and Dušan Georgijević. Interpolation with constraint on frequency region and systems & control application. *Systems & Control Letters*, 97:70–82, 2016.

[8] Kim-Chuan Toh and Lloyd N. Trefethen. The Chebyshev polynomials of a matrix. *SIAM Journal on Matrix Analysis and Applications*, 20(2):400–419, 1998.